AA256 HW2 Solution

Pablo Nunez Martinez

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1 Problem 1

The plane stress and plane strain stiffness matrices for isotropic steel are:

$$[C]plane\ stress\ = \frac{E}{1-\nu^2}\begin{bmatrix}1 & \nu & 0\\ \nu & 1 & 0\\ 0 & 0 & \frac{1-\nu}{2}\end{bmatrix} \quad (1)$$

$$[C]plane\ strain\ = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & 0\\ \nu & 1-\nu & 0\\ 0 & 0 & \frac{1-2\nu}{2}\end{bmatrix} \quad (2)$$

Use engineering strain vectors with ordering (11, 22, 12), where the third strain component is γ_{12}. If the primed frame is rotated by θ from the global frame:

$$\{\sigma'\} = [T]\{\sigma\}, \quad \{e'\} = [T]^{-T}\{e\}$$
$$\{\sigma'\} = [C']\{e'\}$$

Combining these expressions gives the stiffness in the global frame:

$$\{\sigma\} = [T]^{-1}[C'][T]^{-T}\{e\} \quad (3)$$
$$[C] = [T]^{-1}[C'][T]^{-T}$$

The stress transformation matrix is given below, with c = cos θ and s = sin θ:

$$[T] = \begin{bmatrix}c^2 & s^2 & 2cs\\ s^2 & c^2 & -2cs\\ -cs & cs & c^2-s^2\end{bmatrix}$$

To show invariance, it is sufficient to prove that $[T]^{-1}[C'][T]^{-T} = [C']$ for every θ, for each matrix in (1) and (2). Both have the form:

$$[C'] = \begin{bmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & \dfrac{a-b}{2} \end{bmatrix}$$

For plane stress, $a = E/(1-\nu^2)$ and $b = \nu E/(1-\nu^2)$. For plane strain, $a = E(1-\nu)/((1+\nu)(1-2\nu))$ and $b = E\nu/((1+\nu)(1-2\nu))$.

Set $g = (a-b)/2$. Expanding the transformed entries and using $c^2+s^2 = 1$ gives:

$$C_{11} = a(c^4 + s^4) + (2b + 4g)c^2s^2 = a$$
$$C_{22} = a$$
$$C_{12} = b(c^4 + s^4) + (2a - 4g)c^2s^2 = b$$
$$C_{16} = (a - b - 2g)c^3s - (a - b - 2g)cs^3 = 0$$
$$C_{26} = (a - b - 2g)cs^3 - (a - b - 2g)c^3s = 0$$
$$C_{66} = (2a - 2b - 2g)c^2s^2 + g(c^4 + s^4) = g$$

Symmetry supplies the remaining entries. Thus both the plane stress and plane strain stiffness matrices are unchanged by an arbitrary in-plane rotation, as expected for an isotropic material.

2 Problem 2

The local lamina stiffness and compliance relate stress to engineering strain. Let x be the fiber direction and X the global tensile direction. The fiber angle is θ = 20° from X toward Y.

$$\nu_{\gamma x} = \nu_{x\gamma}\frac{E_\gamma}{E_x}$$

$$[S] = \begin{bmatrix} \dfrac{1}{E_x} & \dfrac{-\nu_{x\gamma}}{E_x} & 0 \\ \dfrac{-\nu_{x\gamma}}{E_x} & \dfrac{1}{E_\gamma} & 0 \\ 0 & 0 & \dfrac{1}{G_{x\gamma}} \end{bmatrix}$$

The local stiffness is $[Q] = [S]^{-1}$. Reciprocity gives $\nu_{x\gamma}/E_x = \nu_{\gamma x}/E_\gamma$. Therefore:

$$[Q] = \begin{bmatrix} \frac{E_x}{1-\nu_{x\gamma}\nu_{\gamma x}} & \frac{\nu_{x\gamma}E_\gamma}{1-\nu_{x\gamma}\nu_{\gamma x}} & 0 \\ \frac{\nu_{\gamma x}E_x}{1-\nu_{x\gamma}\nu_{\gamma x}} & \frac{E_\gamma}{1-\nu_{x\gamma}\nu_{\gamma x}} & 0 \\ 0 & 0 & G_{x\gamma} \end{bmatrix}$$

Transforming from local to global coordinates gives:

$$[Q] = [T]^{-1}[Q][T]^{-T} \quad (4)$$

The global compliance can be found either by inverting the global stiffness or by transforming the local compliance:

$$\{e\} = [T]^T[S][T]\{\sigma\} \quad (5)$$

$$[S] = [T]^T[S][T] = [Q]^{-1} \quad (6)$$

$$[T] = \begin{bmatrix} c^2 & s^2 & 2cs \\ s^2 & c^2 & -2cs \\ -cs & cs & c^2 - s^2 \end{bmatrix}, \quad \theta = 20°$$

The two coupon assumptions determine which matrix entry gives the measured tensile modulus. For the long coupon, transverse and shear stresses vanish:

$$\begin{Bmatrix} e_1 \\ e_2 \\ e_6 \end{Bmatrix} = [S] \begin{Bmatrix} \sigma_1 \\ 0 \\ 0 \end{Bmatrix}$$

$$e_1 = S_{11}\sigma_1, \quad E_1 = \frac{1}{S_{11}} \quad (7)$$

For the short coupon, the stated clamping idealization sets transverse and engineering shear strains to zero. Transverse and shear stresses can be nonzero:

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{Bmatrix} = [Q] \begin{Bmatrix} e_1 \\ 0 \\ 0 \end{Bmatrix}$$

$$\sigma_1 = Q_{11}e_1, \quad E_1 * = Q_{11} \quad (8)$$

Using the supplied T800/3900-2 properties, with all moduli converted to psi:

$$[Q] = 10^{6}\begin{bmatrix} 23.302370 & 0.365606 & 0 \\ 0.365606 & 1.305736 & 0 \\ 0 & 0 & 0.900000 \end{bmatrix} psi$$

$$[S] = 10^{-6}\begin{bmatrix} 0.043103 & -0.012069 & 0 \\ -0.012069 & 0.769231 & 0 \\ 0 & 0 & 1.111111 \end{bmatrix} psi^{-1}$$

Substituting θ = 20° into (4) and (6) yields:

$$[Q] = 10^{6}\begin{bmatrix} 18.634771 & 2.460087 & 6.030896 \\ 2.460087 & 1.784373 & 1.038686 \\ 6.030896 & 1.038686 & 2.994481 \end{bmatrix} psi$$

$$[S] = 10^{-6}\begin{bmatrix} 0.156413 & -0.040438 & -0.300989 \\ -0.040438 & 0.712659 & -0.165756 \\ -0.300989 & -0.165756 & 0.997637 \end{bmatrix} psi^{-1}$$

Use full-precision matrices in calculations; the entries displayed here are rounded.

For the long specimen, equation (7) gives:

$$E_1 = \frac{1}{S_{11}} = 6.393340\ msi$$

For the short specimen, equation (8) gives:

$$E_1 * = Q_{11} = 18.634771\ msi$$

The short-coupon apparent modulus is higher because the transverse and shear strains are constrained. The long coupon is free to deform in those directions. The lengths identify the prescribed idealizations; they do not enter these constitutive expressions directly.